

The Spence Loss: Well-Conditioned Value-Function Learning for Reward-Guided Diffusion

David Li-Bland ...

Abstract

Reward-guided diffusion and diffusion fine-tuning are, through entropy-regularized stochastic control, the problem of learning a value function $V(x, t) = \log \mathbb{E}_{\text{base}}[\exp r(X_1) \mid X_t = x]$ —a conditional log-partition function whose gradient is the optimal control. The unbiased Monte-Carlo target for V lives in exponential space, so squared-error regression has gradients that explode for over-predictions and, worse, vanish for under-predictions, where the model is most wrong and gradient clipping cannot help. We observe that every Bregman divergence has the value function as its population minimizer, and introduce the *Spence loss*¹: the Bregman divergence whose gradient has quadratic tails on both sides, with a closed-form numerically-stable gradient. We place it in a family with squared error and the Itakura–Saito divergence. A secondary contribution analyzes why high-dimensional value learning is signal-sparse and studies on-policy importance-sampling schemes. We validate on 2-moons, a dimensional-scaling benchmark designed for clean statistical reads, and GMM-40, where our log-partition estimate is competitive with published diffusion samplers.

- 1 Introduction
- 2 value functions for reward-guided diffusion
- 3 The problem with exponential-space squared error
- 4 Bregman divergences for value learning

4.1 Any Bregman divergence learns the value function

For a strictly convex, differentiable potential φ , the Bregman divergence is

$$D_\varphi(u, v) = \varphi(u) - \varphi(v) - \varphi'(v)(u - v). \quad (1)$$

Squared error is the special case $\varphi(u) = u^2$. The property we exploit:

Lemma 1 (Bregman divergences are proper losses for the value). *Let the regression target be e^q with q a conditionally-unbiased estimate of the value (the H -martingale target, §2). For any strictly convex φ ,*

$$\arg \min_u \mathbb{E}_{q|x_t}[D_\varphi(u, e^q)] = \mathbb{E}[e^q \mid x_t], \quad (2)$$

so the minimizing prediction in log space is $p^ = \log \mathbb{E}[e^q \mid x_t] = V(x_t)$.*

Thus the choice of φ does *not* bias the learned value; it is free to be chosen for *conditioning*. (Proof in App. D.)

4.2 Design criteria and two reference points

We parameterize the prediction in log space, $u = e^p$, and ask how the gradient $\partial L / \partial p$ behaves as the log-error $p - q \rightarrow \pm\infty$.

¹Named for the Spence (dilogarithm) function Li_2 appearing in its value; in our code and experiments it is the **quad** loss.

Table 1: The value-learning loss family. Tails give $\partial L/\partial p$ as the log-error $\rightarrow \pm\infty$. All three share the same minimizer, $V = \log \mathbb{E}[e^q \mid x_t]$ (Lemma 1).

loss	under-predict ($p \rightarrow -\infty$)	over-predict ($p \rightarrow +\infty$)	scale-inv.	uncertainty model
squared error	vanishes (e^{p+q})	explodes (e^{2p})	no	Gaussian (exp)
Itakura–Saito	explodes	linear (bounded)	yes	gamma
Spence (ours)	quadratic ($p e^q$)	quadratic (p)	no	gamma-like

Squared error ($\varphi(u) = u^2$). $\partial L/\partial p = 2e^p(e^p - e^q) = 2e^{2p} - 2e^{p+q}$, which *explodes* as $p \rightarrow +\infty$ and *vanishes* as $p \rightarrow -\infty$ — the under-prediction pathology of §3.

Itakura–Saito ($\varphi(u) = -\ln u$). $D_{\text{IS}}(e^q, e^p) = e^{q-p} - (q - p) - 1$; its gradient in p is $1 - e^{q-p}$, a *linear* (bounded-gradient) tail for over-predictions that removes the vanishing problem, but it still explodes for under-predictions. It is *scale-invariant* ($D_{\text{IS}}(cu, cv) = D_{\text{IS}}(u, v)$) and is the negative log-likelihood of a gamma/exponential observation model, giving a natural uncertainty interpretation for V . A useful midpoint, but one-sided.

4.3 The Spence loss

We propose the Bregman divergence generated by the potential F with

$$F''(u) = -\frac{\ln u}{u(1-u)}, \quad (3)$$

applied to $u = e^p$ against the target e^q . Its gradient with respect to the prediction has the closed form

$$\boxed{\frac{\partial L}{\partial p} = p \frac{e^p - e^q}{e^p - 1}} \quad (4)$$

which is zero iff $p = q$ and, by Lemma 1, has population minimizer V .

Two-sided quadratic tails. From (4),

$$p \rightarrow +\infty : \quad \frac{\partial L}{\partial p} \rightarrow p \quad (\text{no } e^{2p} \text{ blow-up; over-prediction is quadratically penalized}), \quad (5)$$

$$p \rightarrow -\infty : \quad \frac{\partial L}{\partial p} \rightarrow p e^q \quad (\text{non-vanishing corrective signal for under-prediction}). \quad (6)$$

Both tails are linear in p , i.e. the loss is quadratic on both sides — the first loss to fix both pathologies of exponential-space regression at once.

Value via the Spence function. The divergence value (used only for monitoring; the gradient uses (4)) is expressible through the Spence function $S(x) = \text{Li}_2(1 - e^x)$, using the dilogarithm identity $S'(x) = -xe^x/(e^x - 1) = x/\text{expm1}(-x)$; hence the name. Full derivation, the numerically stable branch-wise gradient (finite for all float inputs, unlike the naive autograd which NaNs for $p \gtrsim 88$), and the gamma-like uncertainty interpretation are in App. A.

- 5 the loss in isolation
- 6 Signal sparsity and on-policy learning
- 7 Experiments
- 8 Related work
- 9 Conclusion and limitations

References

- A The Spence loss: derivation and implementation
- B On-policy method zoo
- C Experimental details
- D Additional proofs